

A PROJECT TITLE

ON

Customer Purchase Prediction and Personalized Product Recommendation

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

in

A.Y.: 2026-27

by

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Name of Title: Customer Purchase Prediction and Personalized Product Recommendation

1. Abstract:

The rapid growth of e-commerce platforms has transformed the way people purchase products online. Modern online shopping websites contain thousands of products, making it difficult for customers to discover items that match their interests and preferences. To improve customer satisfaction and increase sales, e-commerce companies use recommendation systems that suggest products based on customer behavior. This project presents a Customer Purchase Prediction and Personalized Product Recommendation System Using Machine Learning.

The main objective of this project is to predict the products that a customer is most likely to purchase and recommend relevant products based on their shopping behavior. The system analyzes customer information such as purchase history, browsing activity, preferred product categories, average spending, product ratings, and frequency of purchases. The collected data is preprocessed by handling missing values, encoding categorical features, scaling numerical features where required, and selecting meaningful features for model training.

Multiple Machine Learning algorithms, including Logistic Regression, Decision Tree, and Random Forest, are implemented and compared to identify the model that provides the highest prediction accuracy. Customer segmentation is performed using K-Means Clustering, which groups customers with similar shopping habits to improve recommendation quality. The trained models are evaluated using Accuracy, Precision, Recall, F1-Score, ROC-AUC, and Cross-Validation to measure their performance and reliability.

2. INTRODUCTION:

E-commerce has become one of the fastest-growing industries worldwide. Every day, millions of customers visit online shopping websites such as Amazon, Flipkart, and Myntra to purchase products. Since these websites contain thousands of products, customers often

find it difficult to discover products that match their interests. As a result, many products remain unnoticed, reducing customer satisfaction and business revenue.

Product recommendation systems have become an important solution to this problem. These systems analyze customer shopping behavior and recommend products that are most relevant to each individual customer. Personalized recommendations improve customer experience, increase product sales, and encourage customers to spend more time on the platform.

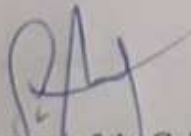
Machine Learning enables computers to identify patterns from historical customer data and make intelligent predictions. Instead of manually recommending products, the system automatically learns customer preferences and predicts future purchases based on previous shopping behavior.

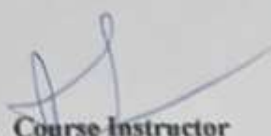
This project proposes a Machine Learning-based recommendation system that predicts customer purchase behavior and recommends products using supervised and unsupervised learning techniques. The project also compares different Machine Learning models to determine the most accurate algorithm for prediction.

3. SOFTWARE REQUIREMENTS:

The following software is required to develop the **Customer Purchase Prediction and Personalized Product Recommendation Using Machine Learning** project:

- **Operating System:** Windows 10/11, Linux, or macOS
- **Programming Language:** Python 3.x
- **IDE/Editor:** Jupyter Notebook, Google Colab, or Visual Studio Code (VS Code)
- **Libraries:**
 - Pandas
 - NumPy
 - Matplotlib
 - Seaborn
 - Scikit-learn (sklearn)
 - Joblib
 - Streamlit
- **Dataset:** E-Commerce Customer Purchase Dataset (CSV format)


Signature of the Guide


Course Instructor